Book 2: Data System Design for AI

Data architecture end to end, grounded in Snowflake and in commercial pharma data. Written for a Manager, Data Specialist who translates business needs into data engineering work and must understand data architectures as a whole: how data is modeled, stored, moved, transformed, governed, served, and made ready for analytics, ML, and generative AI.

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Part I: Foundations of data architecture

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1	`chapters/01-data-architecture-and-the-data-value-chain.md`	Data Architecture and the Data Value Chain: what an architecture is, the layers of a modern data platform, roles (and where the Data Specialist sits), data as a product, reference architectures
2	`chapters/02-data-modeling.md`	Data Modeling: conceptual/logical/physical; normalization; dimensional modeling (facts, dimensions, star schemas, slowly changing dimensions); Data Vault; wide tables for ML; choosing a style
3	`chapters/03-storage-and-compute-warehouses-lakes-lakehouses.md`	Storage and Compute: OLTP vs OLAP, row vs columnar, warehouses vs lakes vs lakehouses; Snowflake architecture deep dive (storage/compute separation, micro-partitions, clustering, virtual warehouses, Time Travel, zero-copy clone, Iceberg tables)
4	`chapters/04-ingestion-and-integration.md`	Ingestion and Integration: batch, CDC, streaming; ETL vs ELT; files and APIs; Snowpipe and Snowpipe Streaming; managed connectors; Snowflake Data Sharing and Marketplace; receiving vendor data (IQVIA, Symphony, Veeva)
5	`chapters/05-transformation-and-orchestration.md`	Transformation and Orchestration: medallion layers, dbt, Snowflake Streams, Tasks and Dynamic Tables, Airflow/Dagster, incremental models, idempotency, backfills, environments and CI/CD
6	`chapters/06-data-quality-and-observability.md`	Data Quality and Observability: quality dimensions, rule design, data contracts, tests in dbt, Snowflake Data Metric Functions, anomaly detection, freshness SLAs, incident handling
7	`chapters/07-governance-security-privacy.md`	Governance, Security, and Privacy: catalogs, lineage, ownership and stewardship, RBAC in Snowflake, masking and row access policies, tags, PII/PHI, HIPAA, GDPR, vendor license restrictions, data sharing agreements
8	`chapters/08-data-products-and-the-semantic-layer.md`	Data Products and the Semantic Layer: what makes a data product, the spec (source-to-target mapping, business rules, data dictionary, DQ rules, SLAs), metrics layers, Snowflake semantic views, serving BI and AI from one definition

Part II: Data foundations for AI

#	File	Chapter
9	`chapters/09-data-for-machine-learning.md`	Data for Machine Learning: feature engineering, feature stores (incl. Snowflake Feature Store), point-in-time correctness, labels, training datasets, train/serve skew, leakage
10	`chapters/10-data-for-generative-ai.md`	Data for Generative AI: unstructured data pipelines, document parsing, embeddings and vector search in Snowflake (Cortex), RAG over governed data, metadata and documentation as AI inputs, text-to-SQL readiness
11	`chapters/11-mlops-and-ai-operations-in-the-warehouse.md`	MLOps and AI Operations in the Warehouse: Snowpark, model registry, batch and near-real-time inference, Cortex functions, monitoring predictions, closing the loop back into the data model
12	`chapters/12-commercial-pharma-data-domain.md`	The Commercial Pharma Data Domain: HCP/HCO master data and MDM, IQVIA (APLD, NPA, NSP, Xponent), Symphony Health, claims (open vs closed), Veeva CRM, affiliations, alignments and territories, market access data, consent and privacy, vendor licensing

Part III: Practice

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13	`chapters/13-requirements-translation-playbook.md`	Requirements Translation Playbook: elicitation, decomposition into epics/stories/tasks, acceptance criteria, UAT, backlog hygiene, communicating status; templates
14	`chapters/14-architecture-patterns-anti-patterns-and-cost.md`	Architecture Patterns, Anti-Patterns, and Cost: data mesh vs centralized, hub-and-spoke, reverse ETL, Snowflake cost management (warehouse sizing, credits, query tuning), migration strategies
15	`chapters/15-case-studies.md`	Case Studies: HCP 360 data product; NBA feature pipeline; Market Access analytics mart; GenAI content sourcing for MLR; Conversational analytics over the commercial mart

Part IV: Agent-ready data platforms

#	File	Chapter
16	`chapters/16-exposing-governed-data-to-agents-on-snowflake.md`	Exposing Governed Data to Agents on Snowflake: Cortex Agents, Snowflake's managed MCP server, semantic views and search as tools, agent service identities, external access, Snowpark Container Services, infrastructure as code, and versioning data-product interfaces

Appendices

File	Contents
`appendix-a-glossary.md`	Glossary of data architecture, Snowflake, and pharma data terms
`appendix-b-templates.md`	Templates: data product spec, source-to-target mapping, data dictionary, DQ rule register, SLA sheet, user story and acceptance criteria
`appendix-c-snowflake-sql-cheat-sheet.md`	Snowflake SQL and feature cheat sheet for specification writing and validation

Reading paths

- **Ship data products now:** 1, 2, 8, 13, 6, then 12 and Appendix B.
- **Understand the platform:** 3, 4, 5, 7, 14.
- **Support AI/ML teams:** 9, 10, 11, 15.
- **Make data available to agents safely:** 7, 8, 10, 16.